Optimal control

Model answers

Question 1

a) The Hamiltonian is

$$H(v(t), u(t), p(t)) = p(t)(u(t) - dv(t)) - (v(t) - v_{\text{ref}})^2$$

in which it is assumed the costate has been normalised so that $p_0 = -1$. The switching function is $\varphi(t) = p^*(t)$ associated with a specific optimal trajectory.

[3 marks]

b) The adjoint equation is

$$\dot{p}^*(t) = -H_v|_* = dp(t) + 2\left(v^*(t) - v_{\text{ref}}\right) \tag{1}$$

The boundary conditions are $v^*(0) = v_0$ and $p^*(t_f) = 0$.

[3 marks]

c) For a singular optimal control to exist, $\varphi(t)$ must identically vanish ($\varphi(t) \equiv 0$) on some time interval $t \in [t_1, t_2]$, that is if there is a time interval $[t_1, t_2]$ during which $\varphi(t) = 0$ for all $t \in [t_1, t_2]$ then $[t_1, t_2]$ is a singular interval. We investigate this possibility by noting that if $\varphi(t) = 0$ for $t \in [t_1, t_2]$ then

$$\varphi(t) = \dot{\varphi}(t) = \ddot{\varphi}(t) = \ldots = 0, \qquad t \in [t_1, t_2]$$

Answers

During a singular interval

$$\varphi(t) = p^*(t) = 0 \tag{2}$$

and by differentiating the switching function and using (1) and (2)

$$\dot{\varphi}(t) = \dot{p}^*(t) = 2\left(v^*(t) - v_{\text{ref}}\right) \tag{3}$$

which means

$$v^*(t) = v_{\text{ref}} \tag{4}$$

makes $\dot{\varphi}(t) = 0$. By differentiating (3) and using the state equation and (4)

$$\ddot{\varphi}(t) = \ddot{p}^*(t) = 2\dot{v}^*(t) = 2\left(u^*(t) - dv_{\text{ref}}\right) \tag{5}$$

from which it can be seen that for the control

$$u^*(t) = dv_{\text{ref}} \tag{6}$$

and when (4) is true, $\dot{\varphi}(t) = 0$, $\ddot{\varphi}(t) = 0$ and all higher-order derivatives of the switching function are obviously 0, hence, $\varphi(t) \equiv 0$ and therefore singular arcs exist. The singular control law is given by (6) at (4), when $dv_{\text{ref}} \leq u_{\max}$ since $|u(t)| \leq u_{\max}$. By considering the maximisation of the Hamiltonian and the singular control law, the optimal control in terms of the switching function is given by

$$u^*(t) = \begin{cases} u_{\max} & \text{if } p(t) > 0 \\ dv_{\text{ref}} & \text{if } p(t) = 0 \\ -u_{\max} & \text{if } p(t) < 0 \end{cases}$$

[6 marks]

d) i) For the case $v_0 > v_{\text{ref}}$ it shouldn't be hard to understand that the optimal policy is to bang (brake with $u^*(t) = -u_{\max}$) to reach v_{ref} as fast as possible and then to stay on the singular arc $u^*(t) = dv_{\text{ref}}$ to maintain $v^*(t) = v_{\text{ref}}$. The state equation for $u^*(t) = -u_{\max}$

$$\dot{v}^*(t) = -u_{\max} - dv^*(t)$$

can be solved to give the state trajectory for the bang segment, as follows:

$$v^*(t) = -\frac{u_{\max}}{d} + \Big(v_0 + \frac{u_{\max}}{d}\Big)e^{-dt} \tag{7}$$

The time, t_s, at which the optimal control switches from the bang to the singular arc can be found by evaluating (7) at $v^*(t) = v_{\text{ref}}$

$$v_{\text{ref}} = -\frac{u_{\max}}{d} + \Big(v_0 + \frac{u_{\max}}{d}\Big)e^{-dt_s} \;\Rightarrow\; t_s = \frac{1}{d}\ln\left(\frac{v_0 + \frac{u_{\max}}{d}}{v_{\text{ref}} + \frac{u_{\max}}{d}}\right)$$

Hence the optimal control is

$$u^*(t) = \begin{cases} -u_{\max} & \text{for } 0 \le t < t_s \\ dv_{\text{ref}} & \text{for } t_s \le t < t_f \end{cases}$$

and the optimal state trajectory is

$$v^*(t) = \begin{cases} -\frac{u_{\max}}{d} + \Big(v_0 + \frac{u_{\max}}{d}\Big)e^{-dt} & \text{for } 0 \le t < t_s \\ v_{\text{ref}} & \text{for } t_s \le t < t_f \end{cases}$$

Answers

If $t_f < t_s$, the singular arc is not reached and the optimal control remains bang over $[0, t_f]$, as

$$u^*(t) = -u_{\max} \quad \text{for } 0 \le t < t_f$$

and the optimal state trajectory is

$$v^*(t) = -\frac{u_{\max}}{d} + \Big(v_0 + \frac{u_{\max}}{d}\Big)e^{-dt} \qquad \text{for } 0 \le t < t_f$$

The above solutions can also more formally be validated by considering solutions of the adjoint equation in (1) for $p(t)$, together with the boundary condition $p^*(t_f) = 0$, although that would require some more tedious calculations.

[4 marks]

ii) For the case $v_0 < v_{\text{ref}}$ it shouldn't be hard to understand that the optimal policy is to bang (accelerate with $u^*(t) = u_{\max}$) to reach v_{ref} as fast as possible and then to stay on the singular arc $u^*(t) = dv_{\text{ref}}$ to maintain $v^*(t) = v_{\text{ref}}$. The state equation for $u^*(t) = u_{\max}$

$$\dot{v}^*(t) = u_{\max} - dv^*(t)$$

can be solved to give the state trajectory for the bang segment, as follows:

$$v^*(t) = \frac{u_{\max}}{d} + \Big(v_0 - \frac{u_{\max}}{d}\Big)e^{-dt} \tag{8}$$

The time, t_s, at which the optimal control switches from the bang to the singular arc can be found by evaluating (8) at $v^*(t) = v_{\text{ref}}$

$$v_{\text{ref}} = \frac{u_{\max}}{d} + \Big(v_0 - \frac{u_{\max}}{d}\Big)e^{-dt_s} \;\Rightarrow\; t_s = \frac{1}{d}\ln\left(\frac{\frac{u_{\max}}{d} - v_0}{\frac{u_{\max}}{d} - v_{\text{ref}}}\right)$$

Hence the optimal control is

$$u^*(t) = \begin{cases} u_{\max} & \text{for } 0 \le t < t_s \\ dv_{\text{ref}} & \text{for } t_s \le t < t_f \end{cases}$$

and the optimal state trajectory is

$$v^*(t) = \begin{cases} \frac{u_{\max}}{d} + \left(v_0 - \frac{u_{\max}}{d}\right)e^{-dt} & \text{for } 0 \le t < t_s \\ v_{\text{ref}} & \text{for } t_s \le t < t_f \end{cases}$$

If $t_f < t_s$, the singular arc is not reached and the optimal control remains bang over $[0, t_f]$, as

$$u^*(t) = u_{\max} \qquad \text{for } 0 \le t < t_f$$

and the optimal state trajectory is

$$v^*(t) = \frac{u_{\max}}{d} + \left(v_0 - \frac{u_{\max}}{d}\right)e^{-dt} \qquad \text{for } 0 \le t < t_f$$

The above solutions can also more formally be validated by considering solutions of the adjoint equation in (1) for $p(t)$, together with the boundary condition $p^*(t_f) = 0$, although that would require some more tedious calculations.

[4 marks]

Answers

Question 2

a) The Hamiltonian is

$$H(x(t), u(t), p(t)) = p(t)\big(-k\,x(t) + u(t)\big) - \big(h\,x(t) + \rho\,u(t)\big)$$
$$= -k\,p(t)\,x(t) - h\,x(t) + \big(p(t) - \rho\big)\,u(t)$$

in which it is assumed that $p(t)$ has been normalised so that $p_0 = -1$.

[2 marks]

b) u must be selected to maximise

$$H(x^*(t), u(t), p^*(t)) = -k\,p^*(t)\,x^*(t) - h\,x^*(t) + \big(p^*(t) - \rho\big)\,u(t)$$

subject to

$$0 \le u(t) \le u_{\max} \qquad \text{for all } t \in [0, t_f]$$

The switching function is $\varphi(t) = p^*(t) - \rho$ associated with a specific optimal trajectory. Therefore,

$$u^*(t) = \begin{cases} u_{\max} & \text{for } \varphi(t) > 0 \ (p^*(t) > \rho) \\ 0 & \text{for } \varphi(t) < 0 \ (p^*(t) < \rho) \\ \tilde{u} & \text{for } \varphi(t) = 0 \ (p^*(t) = \rho) \end{cases}$$

where $\tilde{u}$ is an arbitrary value in $[0, u_{\max}]$. [3 marks]

c) The first boundary condition is $x^*(0) = x_0$ and, since this is a fixed-terminal-time free-endpoint control problem with a terminal cost, the second boundary condition is

$$p^*(t_f) = -\phi_x(x^*(t_f)) = \beta \tag{9}$$

[2 marks]

Answers

d) The adjoint equation is

$$\dot{p}^*(t) = -H_x|_* = kp^*(t) + h$$

which can be solved, also by using (9), and yields

$$p^*(t) = \Big(\beta + \frac{h}{k}\Big)e^{k(t-t_f)} - \frac{h}{k} \tag{10}$$

[3 marks]

e) By evaluating $\dot{\varphi}(t)$ and substituting from (10)

$$\dot{\varphi}(t) = \dot{p}^*(t) = kp^*(t) + h = k\Big(\beta + \frac{h}{k}\Big)e^{k(t-t_f)}$$

and, therefore, $\dot{\varphi}(t) > 0$. Hence $\varphi(t)$ is strictly increasing, therefore, there is at most one switching time $t_s \in (0, t_f)$, with direction from $u^*(t) = 0$ to $u^*(t) = u_{\max}$. The switching time, if it exists, satisfies $\varphi(t_s) = 0 \iff p^*(t_s) = \rho$, which gives

$$\Big(\beta + \frac{h}{k}\Big)e^{k\,(t_s - t_f)} - \frac{h}{k} = \rho$$

and therefore

$$t_s = t_f - \frac{1}{k}\ln\left(\frac{\beta + \frac{h}{k}}{\rho + \frac{h}{k}}\right) \tag{11}$$

[4 marks]

f) For $t_s \in (0, t_f)$, the second term on the right hand side of (11) should satisfy

$$0 < \frac{1}{k} \ln\left(\frac{\beta + \frac{h}{k}}{\rho + \frac{h}{k}}\right) < t_f$$

There are, therefore, three possible solution cases:

- $\frac{1}{k} \ln\left(\frac{\beta+\frac{h}{k}}{\rho+\frac{h}{k}}\right) \leq 0$, which occurs when $\beta \leq \rho$. This implies

  $$\varphi(t_f) = \beta - \rho \leq 0$$

  which means $u^*(t_f) = 0$, with no switch possible at earlier times. The optimal control is, therefore, always-off: $u^*(t) = 0$ for $0 \leq t \leq t_f$.

- $\frac{1}{k} \ln\left(\frac{\beta+\frac{h}{k}}{\rho+\frac{h}{k}}\right) \geq t_f$, which occurs when $\beta \geq e^{kt_f}\left(\rho + \frac{h}{k}\right) - \frac{h}{k}$. This implies

  $$\varphi(0) = \left(\beta + \frac{h}{k}\right)e^{-kt_f} - \left(\frac{h}{k} + \rho\right) \geq 0$$

  which means $u^*(0) = u_{\max}$, with no switch possible at later times. The optimal control is, therefore, always-on: $u^*(t) = u_{\max}$ for $0 \leq t \leq t_f$.

- $0 < \frac{1}{k} \ln\left(\frac{\beta+\frac{h}{k}}{\rho+\frac{h}{k}}\right) < t_f$, which occurs when

  $$\rho < \beta < e^{kt_f}\left(\rho + \frac{h}{k}\right) - \frac{h}{k}$$

  In this case there is a single switch at t_s:

  $$u^*(t) = \begin{cases} 0 & \text{for } 0 \leq t < t_s \\ u_{\max} & \text{for } t_s \leq t \leq t_f \end{cases}$$

Answers

[6 marks]

Question 3

a) The Lagrangian is

$$L = \frac{(y'(x) - x)^2}{y'(x)}$$

[2 marks]

b) Since the Lagrangian does not depend explicitly on y, but only on its derivative, one can make use of the "no y" result of the Euler-Lagrange equation, as follows:

$$L_{y'} = c_0, \text{ with } c_0 \in \mathbb{R},$$

where $L_{y'}$ can be calculated as

$$L_{y'} = 1 - \frac{x^2}{(y'(x))^2} \tag{12}$$

Therefore, the Euler-Lagrange equation results in:

$$1 - \frac{x^2}{(y'(x))^2} = c_0$$

which can be rewritten as

$$y'(x) = 2c_1 x \tag{13}$$

with $2c_1 = \pm\frac{1}{\sqrt{1-c_0}}$.

[4 marks]

Answers

c) (13) can be integrated as

$$y(x) = c_1 x^2 + c_2, \text{ with } c_2 \in \mathbb{R}$$

The extremal subarcs have the form given in the equation above, but the constants of integration will generally be different on the two sides of the corner point at $c \in [-1, 1]$. Therefore the two subarcs are:

$$y(x) = c_1 x^2 + c_2 \qquad \text{for } x \in [-1, c] \tag{14}$$
$$y(x) = c_3 x^2 + c_4 \qquad \text{for } x \in [c, 1] \tag{15}$$

which give the general form of the extremal curve.

[3 marks]

d) The first Weierstrass-Erdmann corner condition for one corner point at $c \in [-1, 1]$ is

$$L_{y'}(c, y(c), y'(c^-)) = L_{y'}(c, y(c), y'(c^+))$$

which gives from (12) that

$$1 - \frac{c^2}{(y'(c^-))^2} = 1 - \frac{c^2}{(y'(c^+))^2} \tag{16}$$

The second Weierstrass-Erdmann corner condition is

$$L_{y'}(c, y(c), y'(c^-))y'(c^-) - L(c, y(c), y'(c^-)) = L_{y'}(c, y(c), y'(c^+))y'(c^+) - L(c, y(c), y'(c^+))$$

which gives

$$2\left(-\frac{c^2}{y'(c^-)}+c\right)=2\left(-\frac{c^2}{y'(c^+)}+c\right) \tag{17}$$

For $c\neq 0$, (16) is satisfied by $|y'(c^-)|=|y'(c^+)|$ and (17) further forces $y'(c^-)=y'(c^+)$, which means there is no genuine corner at $c\neq 0$. On the contrary, at $c=0$ there is continuity of $L_{y'}$ and $L_{y'}y'-L$ $(=0)$ without requiring $y'(c^-)=y'(c^+)$, therefore, a corner is admissible only at $x=0$.

[4 marks]

e) Unless $c_1=-c_3$ in (14) and (15), a small shift of the corner would move "length" from the more expensive side to the cheaper side, reducing $J(y)$, thus contradicting optimality. Hence, the free corner forces

$$c_1=-c_3 \tag{18}$$

which is the equal and opposite $y''(x)$ argument. Furthermore, the extremals must satisfy the boundary conditions, $y(-1)=0$ and $y(1)=1$, therefore

$$c_1+c_2=0 \tag{19}$$
$$c_3+c_4=1 \tag{20}$$

Additionally, at the corner point $c=0$ it must also be true that

$$y(c^-)=y(c^+)$$

and therefore by evaluating (14) and (15) gives

$$c_2=c_4 \tag{21}$$

Answers

The equations in (18), (19), (20), and (21) can be used to calculate the constant parameters as $c_1=-\frac{1}{2}$ and $c_2=c_3=c_4=\frac{1}{2}$. Therefore, the extremal curve is

$$y(x)=\begin{cases}-\dfrac{x^2}{2}+\dfrac{1}{2} & \text{for } x\in[-1,0]\\[2ex] \dfrac{x^2}{2}+\dfrac{1}{2} & \text{for } x\in[0,1]\end{cases} \tag{22}$$

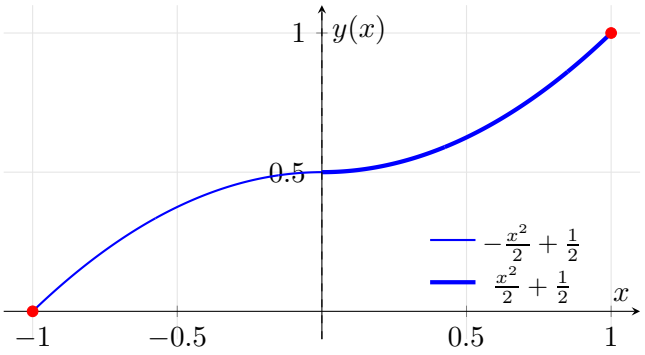


Figure 1: Stationary broken extremal with a free corner at $x=0$.

[7 marks]

Question 4

a) The Hamilton-Jacobi-Bellman (HJB) differential equation for this problem is

$$-V_t(t,x) = \inf_{u\in[0,u_{\max}]} \Big\{ hx(t) + \rho\, u(t) + V_x(t,x)\,(-k\,x(t) + u(t)) \Big\}$$

[2 marks]

b) The expression inside the infimum can be rewritten as

$$hx(t) - V_x(t,x)kx(t) + (\rho + V_x(t,x))u(t)$$

therefore the infimum is achieved by setting

$$u(t) = \begin{cases} u_{\max} & \text{for } \rho + V_x(t,x) < 0 \\ 0 & \text{for } \rho + V_x(t,x) > 0 \\ \tilde{u} & \text{for } \rho + V_x(t,x) = 0 \end{cases} \tag{23}$$

where $\tilde{u}$ is an arbitrary value in $[0, u_{\max}]$. By substituting the control (23) into the HJB equation, one obtains

$$\begin{cases} -V_t(t,x) = hx(t) - V_x(t,x)kx(t) + (\rho + V_x(t,x))u_{\max} & \text{for } \rho + V_x(t,x) < 0 \\ -V_t(t,x) = hx(t) - V_x(t,x)kx(t) & \text{for } \rho + V_x(t,x) \geq 0 \end{cases} \tag{24}$$

Answers

[3 marks]

c) Based on the terminal cost of $-\beta x(t_f)$, the boundary condition that must be satisfied by the value function is

$$V(t_f, x) = -\beta x \qquad \text{for all } x$$

[2 marks]

d) i) In order for the assumed solution to match the boundary condition in part c), make

$$a(t_f) = -\beta, \qquad b(t_f) = 0 \tag{25}$$

[2 marks]

ii) By differentiating the assumed solution $V(t,x) = a(t)x(t) + b(t)$ one obtains

$$V_x(t,x) = a(t) \tag{26}$$

and

$$V_t(t,x) = \dot{a}(t)x(t) + \dot{b}(t)$$

Then by substituting these terms in the HJB equation (24), yields

$$\begin{cases} -\dot{a}(t)x(t) - \dot{b}(t) = hx(t) - a(t)kx(t) + (\rho + a(t))u_{\max} & \text{for } \rho + a(t) < 0 \\ -\dot{a}(t)x(t) - \dot{b}(t) = hx(t) - a(t)kx(t) & \text{for } \rho + a(t) \geq 0 \end{cases}$$

Since this equation must be satisfied for all $x(t)$, it gives the differential equation for $a(t)$

$$\dot{a}(t) - ka(t) = -h \tag{27}$$

and the differential equation for $b(t)$

$$\begin{cases} -\dot{b}(t) = (\rho + a(t))u_{\max} & \text{for } \rho + a(t) < 0 \\ -\dot{b}(t) = 0 & \text{for } \rho + a(t) \geq 0 \end{cases}$$

[5 marks]

iii) In light of (26), (23) depends only on $a(t)$ as

$$u(t) = \begin{cases} u_{\max} & \text{for } \rho + a(t) < 0 \\ 0 & \text{for } \rho + a(t) > 0 \\ \tilde{u} & \text{for } \rho + a(t) = 0 \end{cases} \tag{28}$$

$a(t)$ can be obtained by solving (27), while also using (25), to give

$$a(t) = -\left(\beta + \frac{h}{k}\right) e^{k(t-t_f)} + \frac{h}{k} \tag{29}$$

The (unique) switching time, t_s, if it lies in $(0, t_f)$, solves $\rho + a(t_s) = 0$:

$$\left(\beta + \frac{h}{k}\right) e^{k(t_s - t_f)} = \rho + \frac{h}{k}$$

and therefore

$$t_s = t_f - \frac{1}{k} \ln\left(\frac{\beta + \frac{h}{k}}{\rho + \frac{h}{k}}\right)$$

There are, therefore, three possible solution cases:

- $t_s \geq t_f \Rightarrow$ no switch: This occurs when $\frac{1}{k}\ln\left(\frac{\beta+\frac{h}{k}}{\rho+\frac{h}{k}}\right) \leq 0$, which occurs when $\beta \leq \rho$. This implies

  $$\rho + a(t_f) = \rho - \beta \geq 0$$

  which means $u(t_f) = 0$ and the optimal control is, therefore, always-off: $u(t) = 0$ for $0 \leq t \leq t_f$.

- $t_s \leq 0 \Rightarrow$ no switch: This occurs when $\frac{1}{k}\ln\left(\frac{\beta+\frac{h}{k}}{\rho+\frac{h}{k}}\right) \geq t_f$, which occurs when $\beta \geq e^{kt_f}\left(\rho + \frac{h}{k}\right) - \frac{h}{k}$. This implies

  $$\rho + a(0) = -\left(\beta + \frac{h}{k}\right) e^{-kt_f} + \left(\frac{h}{k} + \rho\right) \leq 0$$

  which means $u(0) = u_{\max}$, and the optimal control is, therefore, always-on: $u(t) = u_{\max}$ for $0 \leq t \leq t_f$.

- $0 < t_s < t_f$, which occurs when $0 < \frac{1}{k}\ln\left(\frac{\beta+\frac{h}{k}}{\rho+\frac{h}{k}}\right) < t_f$, which occurs when

  $$\rho < \beta < e^{kt_f}\left(\rho + \frac{h}{k}\right) - \frac{h}{k}$$

  In this case there is a single switch at t_s from $u(t) = 0$ to $u(t) = u_{\max}$ (because $\rho + a(t)$ decreases monotonically with time):

  $$u(t) = \begin{cases} 0 & \text{for } 0 \leq t < t_s \\ u_{\max} & \text{for } t_s \leq t \leq t_f \end{cases}$$

[6 marks]

Answers